

NEWCASTLE NOBBYS HEAD, Australia

WMO: 947740

Lat: 32.919S

Lon: 151.798E

Elev: 33

StdP: 100.93

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	8.6	-0.3	3.7	14.7	0.9	4.1	14.2	19.6	16.0	16.3	14.7	5.5	300	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.1	31.0	19.6	28.1	19.5	26.1	20.4	23.5	25.7	22.9	25.0	22.3	24.4	8.7	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	17.7	24.4	22.2	17.0	23.9	21.7	16.4	23.5	70.2	25.8	68.0	25.2	65.9	24.4	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.8	13.9	12.3	DB	5.6	37.3	1.1	2.3	4.8	39.0	4.2	40.3	3.5	41.7	2.7	43.4	(4)
(5)			WB	3.8	24.6	0.9	1.0	3.2	25.3	2.6	25.8	2.1	26.4	1.5	27.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.4	22.5	22.5	21.8	19.4	16.6	14.4	13.5	14.4	16.6	18.4	19.9	21.3	(6)	
	DBStd	3.82	2.07	1.84	1.84	1.96	1.98	1.83	1.75	2.21	2.56	2.71	2.52	2.19	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	573	0	0	1	10	60	119	149	126	66	32	9	2	(9)	
	CDD10.0	3071	388	351	366	283	205	131	110	135	197	259	296	350	(10)	
	CDD18.3	602	129	118	108	43	7	0	0	2	13	33	55	94	(11)	
	CDH23.3	1627	375	299	205	71	9	0	1	11	78	159	191	229	(12)	
	CDH26.7	409	107	47	32	11	0	0	0	1	17	55	70	70	(13)	
(14) Wind	WSAvg	6.0	6.3	6.2	5.9	5.5	5.5	5.9	5.7	5.9	6.1	6.2	6.4	6.3	(14)	
(15) Precipitation	PrecAvg	1130	101	107	126	105	122	140	68	68	60	78	83	83	(15)	
	PrecMax	1801	334	559	337	338	441	408	269	224	196	185	204	225	(16)	
	PrecMin	597	2	10	3	4	9	22	0	1	2	5	12	7	(17)	
	PrecStd	261	74	94	81	88	86	87	53	55	44	47	55	60	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.8	32.2	30.4	28.3	24.6	21.4	21.9	25.3	29.1	31.6	33.1	33.5	(19)	
		MCWB	20.9	22.4	21.2	18.8	16.9	15.9	14.2	15.6	16.4	17.3	18.8	19.9	(20)	
	2%	DB	29.1	27.4	26.5	25.2	22.5	19.5	19.5	21.8	25.6	27.8	28.0	28.1	(21)	
		MCWB	21.1	22.6	21.3	18.9	16.8	14.8	13.1	13.5	15.6	17.3	18.5	19.5	(22)	
	5%	DB	25.9	25.7	25.0	23.6	21.2	18.5	18.1	19.8	22.9	24.3	24.7	25.3	(23)	
		MCWB	21.6	22.2	21.4	19.0	16.9	14.4	12.7	13.1	14.7	16.9	19.0	20.5	(24)	
	10%	DB	24.8	24.9	24.2	22.6	20.2	17.7	17.0	18.4	20.6	21.9	22.8	23.9	(25)	
		MCWB	21.4	21.8	21.0	18.8	16.4	14.1	12.4	13.0	14.9	17.3	19.3	20.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.8	24.6	24.0	21.9	19.9	17.2	16.6	17.0	18.9	20.5	22.0	23.4	(27)	
		MCDB	27.3	26.7	25.8	23.8	21.2	19.2	18.8	20.5	21.7	23.6	25.0	25.8	(28)	
	2%	WB	23.1	23.7	22.9	20.8	18.9	16.5	15.2	16.0	18.0	19.6	21.2	22.3	(29)	
		MCDB	25.5	25.6	24.7	22.9	20.6	18.2	17.2	18.9	20.7	22.2	23.8	24.5	(30)	
	5%	WB	22.6	23.0	22.2	20.2	18.0	15.8	14.2	15.1	17.2	18.9	20.6	21.6	(31)	
		MCDB	24.8	25.2	24.1	22.5	20.2	17.5	16.5	18.2	20.2	21.5	23.0	23.8	(32)	
	10%	WB	22.1	22.4	21.6	19.6	17.2	15.1	13.5	14.2	16.4	18.2	19.9	21.0	(33)	
		MCDB	24.3	24.4	23.7	21.9	19.5	16.9	15.9	17.3	19.5	21.1	22.2	23.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.3	4.1	4.6	5.8	6.7	6.2	7.1	7.5	7.2	6.2	5.2	4.7	(35)	
		MCDBR	7.5	6.1	6.4	8.1	8.4	7.3	9.1	10.2	11.8	12.0	10.3	8.6	(36)	
	5% WB	MCWBR	3.1	3.0	3.3	4.2	4.3	4.1	4.8	5.0	5.1	4.7	3.8	3.6	(37)	
		MCWBR	5.6	5.1	5.4	6.4	6.4	6.0	6.7	7.8	8.2	7.1	6.6	6.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.395	0.386	0.355	0.333	0.316	0.310	0.300	0.309	0.344	0.363	0.386	0.383	(40)	
		taud	2.455	2.504	2.582	2.616	2.616	2.625	2.638	2.601	2.497	2.472	2.450	2.477	(41)	
	Ebn at Noon		944	932	928	896	862	841	872	909	923	944	946	959	(42)	
		Edh at Noon	120	110	95	82	73	68	70	82	102	112	119	118	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.60	5.81	4.89	3.91	3.07	2.45	2.86	3.79	4.98	5.75	6.19	6.66	(44)	
	RadStd		0.42	0.48	0.42	0.31	0.23	0.20	0.23	0.23	0.26	0.46	0.57	0.52	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)		+0.43	N/A	N/A	+0.77	N/A	N/A	-1	-85	+190	+103

Nomenclature: See separate page